

Worksheet 1: Fractions — Solutions

AIML Foundations Mathematics

Dublin and Dún Laoghaire ETB

Instructor: Josh Aaron

Instructor Copy — Not for Distribution

Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

(Answers on separate page)

Part A

1. $\frac{19}{12}$
2. $\frac{5}{24}$
3. $\frac{1}{6}$
4. 6
5. $\frac{3}{4}$
6. $\frac{2}{3}$
7. $\frac{4}{3}$
8. $\frac{22}{15}$
9. $\frac{7}{2}$ or $3\frac{1}{2}$
10. 4

Part B

11. $\frac{2x}{3}$
12. $\frac{3a^2}{5b^2}$
13. $x - 2$
14. $\frac{x-3}{x+3}$
15. 2
16. $\frac{x-3}{x+2}$
17. $\frac{3x-2}{x+2}$
18. $\frac{x^2+2x+4}{x+2}$

Part C

19. $\frac{5}{x}$
20. $\frac{3}{x+1}$
21. $\frac{3y+2x}{xy}$
22. $\frac{x^2+4}{(x+2)(x-2)}$
23. $\frac{2}{x^2-1}$
24. $\frac{2x+1}{x^2-1}$
25. $\frac{x-3}{2x}$
26. $x(x+2)$ or x^2+2x

Part D

27. $\frac{y+x}{y-x}$
28. $\frac{x-1}{3(x+1)}$
29. $\frac{x-1}{x}$
30. $\frac{a^2+b^2}{a^2-b^2}$

Part E (Brief answers)

31. (a) Increases without bound (b) Approaches 0
32. (a) Approaches 0 (b) Increases without bound
33. (a) $1 + \frac{1}{x}$ (b) Approaches 1 (c) $\frac{1}{x}$ becomes negligibly small
34. (a) $\frac{1}{2}$, $\frac{100}{101}$, $\frac{10000}{10001}$ (b) Approaches 1 (c) Never reaches 1; denominator always larger
35. (a) Stays the same (b) Depends—increases if $\frac{a}{b} < 1$, decreases if $\frac{a}{b} > 1$ (c) Adding 1 to both pulls the fraction toward 1
36. (a) $\frac{x}{x+1}$ (b) Approaches 1 (c) Approaches 0

Part F

37. $\frac{x+1}{x+2}$
38. $\frac{3x^2+6x+2}{x(x+1)(x+2)}$
39. $c = \frac{ab}{a+b}$
40. (a) $x + \frac{1}{x} = \frac{5}{2}$ (b) $2x^2 - 5x + 2 = 0$ (c) $x = 2$ or $x = \frac{1}{2}$
